A Comprehensive Data Corpus of the Linear A Signary for Computational Analysis

Section 1: Introduction to the Linear A Corpus and Signary

1.1 Defining the Linear A Script and Corpus

Linear A is an undeciphered writing system employed by the Minoan civilization of Crete and the surrounding Aegean islands from approximately 1800 to 1450 BCE. It stands as one of the four principal scripts of the second millennium BC Aegean, alongside its descendant Linear B, the related Cypro-Minoan syllabary, and the pictographic Cretan Hieroglyphic script. The discovery of Linear A is credited to Sir Arthur Evans in 1900, who termed it "linear" due to its abstract characters formed from lines inscribed in clay, distinguishing it from the more pictorial hieroglyphs.

While Linear A is the direct antecedent of Linear B—the script used by the Mycenaeans to write an early form of Greek and famously deciphered in 1952—its underlying language, termed "Minoan," remains unknown. Attempts to apply the phonetic values of Linear B to homomorphic (similarly shaped) Linear A signs do not yield an intelligible language, confirming that Minoan is not a Hellenic dialect.

The extant corpus of Linear A is notably small, comprising approximately 1,427 artifacts which bear a total of 7,150 to 7,400 sign occurrences. This limited dataset presents a significant challenge for any analytical endeavor, particularly when contrasted with the more than 57,000 sign instances available for Linear B. The brevity of the inscriptions, which are predominantly administrative lists, further complicates decipherment efforts. The foundational reference for this corpus is the five-volume *Recueil des inscriptions en linéaire A* (GORILA), compiled by Louis Godart and Jean-Pierre Olivier, which remains the authoritative print collection. Modern computational research builds upon this work, leveraging crucial digital resources such as John Younger's *Linear A Texts in phonetic transcription* and the SigLA palaeographical database, which provide structured access to the corpus.

1.2 The Structure of the Signary

The complete inventory of Linear A signs is a composite of several distinct functional categories, with scholarly estimates of the total count varying. A comprehensive aggregation suggests a signary of approximately 374 to 411 distinct signs, a number derived from the summation of simple signs, complex signs (ligatures), and fractional signs. This inventory can be broken down as follows:

- **Simple Signs:** Approximately 180 graphically independent signs that primarily function as phonetic syllabograms but can also be used as logograms. Within this group, a "core" of about 90 signs appears with notable frequency across the script's chronological and

geographic range.

- **Complex/Composite Signs:** Approximately 164 signs formed by the ligature or juxtaposition of two or more simple signs. These function almost exclusively as logograms and represent a distinctive feature of Linear A that was not adopted into Linear B.
- **Fractional Signs:** A specialized set of 30 to 47 signs, known as klasmatograms, used to denote fractional values.

The total size of this signary, comfortably exceeding 256 signs, provides the explicit justification for scaling the Octeract Cypher framework beyond its default 8-bit operational environment. As stipulated by the framework's own scalability protocol, a signary of this magnitude necessitates a higher-dimensional space to ensure a unique vector can be assigned to each sign without compromising analytical resolution. The transition to a 9-bit (Enneact) mode, with its 512-vertex capacity, is therefore a mandatory and mathematically sound step for the ingestion of the complete Linear A corpus.

A significant characteristic of the Linear A signary is the prevalence of data sparsity. A large proportion of the inventory, particularly the complex logograms, are *hapax legomena*—signs that are attested only once in the entire corpus. This creates a "long-tail" frequency distribution where a few signs are very common, while a vast number are exceptionally rare. This statistical property is not merely an artifact of an incomplete archaeological record; it suggests that the script possessed a highly specialized and context-specific function, likely for recording unique or infrequent commodities and transactions. For any computational model, this sparsity presents a fundamental challenge, as patterns for low-frequency signs are difficult to learn and generalize. The analytical framework must therefore be inherently robust to this feature, employing methods capable of handling sparse data effectively.

1.3 Sign Classification System

To structure the corpus for analysis, all signs are categorized according to their primary function, based on established academic conventions. However, it is crucial to recognize that these categories are not always mutually exclusive. The function of many signs is context-dependent; a simple sign may act as a phonetic syllabogram when part of a multi-sign group but function as a logogram when it appears in isolation before a numeral. This functional duality is a key grammatical feature of the script. A static classification is an oversimplification, and a robust analysis must account for the sign's position and surrounding context to determine its function in any given instance. The following classifications represent the primary or most common role of each sign:

- **Syllabogram:** A phonetic sign representing an open syllable (e.g., Vowel, Consonant-Vowel).
- **Logogram (Simple):** A simple sign used ideographically to represent a commodity, person, or concept.
- **Logogram (Complex):** A composite sign (ligature) used ideographically.
- **Numeral:** A sign representing an integer value.
- **Fraction:** A sign representing a fractional value.

Section 2: The Complete Signary: Master List and Frequencies

This section presents the master data table for the complete Linear A signary, designed for

direct computational ingestion. The list is compiled from the most comprehensive available academic resources, primarily the GORILA corpus and the Mnamon digital database, which provides a standardized numbering system. The total sign count in this synthesized list is 374. Frequencies are provided as best-effort estimates, combining quantitative data from specific studies with qualitative tiers derived from scholarly consensus, reflecting the current state of research.

Table 1: Master Sign List, Classification, and Corpus Frequency of the Linear A Signary

Sign_ID	Glyph	Sign_Category	Total_Frequency	Notes
Simple Signs (Syllabograms & Simple Logograms)				
AB01	□	Syllabogram	High (>100)	Homomorph: LB *da*.
AB02	□	Syllabogram	High (>100)	Homomorph: LB *ro*.
AB03	□	Syllabogram	High (>100)	Homomorph: LB *pa*.
AB04	□	Syllabogram	High (>100)	Homomorph: LB *te*.
AB05	□	Syllabogram	Medium (10-99)	Homomorph: LB *to*.
AB06	□	Syllabogram	High (>100)	Homomorph: LB *na*.
AB07	□	Syllabogram	High (>100)	Homomorph: LB *di*.
AB08	□	Syllabogram	High (>100)	Homomorph: LB *a*. Core vowel.
AB09	□	Syllabogram	High (>100)	Homomorph: LB *se*.
AB10	□	Syllabogram	High (>100)	Homomorph: LB *u*. Core vowel.
AB11	□	Syllabogram	Medium (10-99)	Homomorph: LB *po*.
AB13	□	Syllabogram	Medium (10-99)	Homomorph: LB *me*.
AB16	□	Syllabogram	Medium (10-99)	Homomorph: LB *qa*.
AB17	□	Syllabogram	Low (2-9)	Homomorph: LB *za*.
AB20	□	Syllabogram	Low (2-9)	Homomorph: LB *zo*.
AB21	□	Logogram (Simple)	High (>100)	OVIS (sheep). Also syllabic *qi*.
AB21f	□	Logogram (Simple)	Medium (10-99)	OVISf (ewe).

Sign_ID	Glyph	Sign_Category	Total_Frequency	Notes
AB21m	□	Logogram (Simple)	Medium (10-99)	OVISm (ram).
AB22	□	Logogram (Simple)	Medium (10-99)	CAP (goat). Also syllabic *pi*$_2$.
AB22f	□	Logogram (Simple)	Medium (10-99)	CAPf (she-goat).
AB22m	□	Logogram (Simple)	Medium (10-99)	CAPm (he-goat).
AB23	□	Logogram (Simple)	Medium (10-99)	BOS (bovine). Also syllabic *mu*.
AB23m	□	Logogram (Simple)	Low (2-9)	BOSm (ox/bull).
AB24	□	Syllabogram	High (>100)	Homomorph: LB *ne*.
AB26	□	Syllabogram	High (>100)	Homomorph: LB *ru*.
AB27	□	Syllabogram	High (>100)	Homomorph: LB *re*.
AB28	□	Syllabogram	High (>100)	Homomorph: LB *i*. Core vowel.
A28b	□	Syllabogram	Low (2-9)	Variant of AB28.
AB29	□	Syllabogram	Low (2-9)	Homomorph: LB *pu*$_2$.
AB30	□	Logogram (Simple)	High (>100)	FIC (figs). Also syllabic *ni*.
AB31	□	Syllabogram	High (>100)	Homomorph: LB *sa*.
AB34	□	Syllabogram	Hapax (1)	
AB37	□	Syllabogram	High (>100)	Homomorph: LB *ti*.
AB38	□	Syllabogram	High (>100)	Homomorph: LB *e*. Core vowel.
AB39	□	Syllabogram	High (>100)	Homomorph: LB *pi*.
AB40	□	Syllabogram	Medium (10-99)	Homomorph: LB *wi*.
AB41	□	Syllabogram	High (>100)	Homomorph: LB *si*.
AB44	□	Syllabogram	Medium (10-99)	Homomorph: LB *ke*.
AB45	□	Syllabogram	High (>100)	Homomorph: LB *de*.
AB46	□	Syllabogram	Medium (10-99)	Homomorph: LB *je*.
AB47	□	Syllabogram	Low (2-9)	
AB48	□	Syllabogram	Low (2-9)	Homomorph: LB

Sign_ID	Glyph	Sign_Category	Total_Frequency	Notes
				nwa.
AB49	□	Syllabogram	Low (2-9)	
AB50	□	Syllabogram	High (>100)	Homomorph: LB *pu*.
AB51	□	Syllabogram	High (>100)	Homomorph: LB *du*.
AB53	□	Syllabogram	High (>100)	Homomorph: LB *ri*.
AB54	□	Logogram (Simple)	Medium (10-99)	TELA (cloth). Also syllabic *wa*.
AB55	□	Syllabogram	High (>100)	Homomorph: LB *nu*.
AB56	□	Syllabogram	Low (2-9)	Homomorph: LB pa_2.
AB57	□	Syllabogram	High (>100)	Homomorph: LB *ja*.
AB58	□	Syllabogram	High (>100)	Homomorph: LB *su*.
AB59	□	Syllabogram	High (>100)	Homomorph: LB *ta*.
AB60	□	Syllabogram	High (>100)	Homomorph: LB *ra*.
AB61	□	Syllabogram	Medium (10-99)	Homomorph: LB *o*. Core vowel.
AB65	□	Syllabogram	Low (2-9)	Homomorph: LB *ju*.
AB66	□	Syllabogram	Low (2-9)	Homomorph: LB ta_2.
AB67	□	Syllabogram	High (>100)	Homomorph: LB *ki*.
AB69	□	Syllabogram	High (>100)	Homomorph: LB *tu*.
AB70	□	Syllabogram	Medium (10-99)	Homomorph: LB *ko*.
AB73	□	Syllabogram	High (>100)	Homomorph: LB *mi*.
AB74	□	Syllabogram	Medium (10-99)	Homomorph: LB *ze*.
AB76	□	Syllabogram	High (>100)	Homomorph: LB ra_2.
AB77	□	Syllabogram	High (>100)	Homomorph: LB *ka*.
AB78	□	Syllabogram	Medium (10-99)	Homomorph: LB *qe*.
AB79	□	Syllabogram	High (>100)	Homomorph: LB *zu*.

Sign_ID	Glyph	Sign_Category	Total_Frequency	Notes
AB80	□	Syllabogram	High (>100)	Homomorph: LB *ma*.
AB81	□	Syllabogram	High (>100)	Homomorph: LB *ku*.
AB82	□	Logogram (Simple)	Low (2-9)	
AB85	□	Logogram (Simple)	Medium (10-99)	SUS (pig). Also syllabic *au*.
AB86	□	Syllabogram	Low (2-9)	
AB87	□	Syllabogram	Low (2-9)	Homomorph: LB *twe*.
A100/102	□	Logogram (Simple)	High (>100)	VIR (man/person).
AB118	□	Syllabogram	Medium (10-99)	
AB120	□	Logogram (Simple)	High (>100)	GRA (grain/wheat).
A120b	□	Logogram (Simple)	Medium (10-99)	Variant of GRA.
AB122	□	Logogram (Simple)	High (>100)	OLIV (olives).
AB123	□	Logogram (Simple)	Low (2-9)	
AB131a	□	Logogram (Simple)	High (>100)	VINa (wine).
AB131b	□	Logogram (Simple)	Medium (10-99)	VINb (wine).
A131c	□	Logogram (Simple)	Medium (10-99)	VINc (wine).
AB164	□	Logogram (Simple)	Low (2-9)	
AB171	□	Syllabogram	Low (2-9)	
AB180	□	Syllabogram	Low (2-9)	
AB188	□	Syllabogram	Hapax (1)	
AB191	□	Logogram (Simple)	Hapax (1)	GAL (helmet).
A301	□	Syllabogram	High (>100)	
A302	□	Logogram (Simple)	High (>100)	OLE (oil).
A303	□	Logogram (Simple)	High (>100)	CYP (cyperus).
A304	□	Logogram (Simple)	Medium (10-99)	
A305	□	Logogram (Simple)	Medium (10-99)	

Sign_ID	Glyph	Sign_Category	Total_Frequency	Notes
A306	□	Syllabogram	Low (2-9)	
...	...	...	...	*Additional 44 simple signs (A307-A371) exist, mostly hapax or low frequency.*
Complex Signs (Logograms)				
A501	□	Logogram (Complex)	Hapax (1)	Ligature.
A502	□	Logogram (Complex)	Hapax (1)	Ligature.
A503	□	Logogram (Complex)	Hapax (1)	Ligature.
...	...	...	...	*Total of 164 complex signs (A501-A664), overwhelmingly hapax legomena.*
Numerical Signs				
1	'	Numeral	High (>100)	Unit (1).
10	-	Numeral	High (>100)	Ten (10).
100	∘	Numeral	High (>100)	Hundred (100).
1000	⌖	Numeral	High (>100)	Thousand (1000).
Fractional Signs				
A701 (A)	□	Fraction	Medium (10-99)	Hypothesized value varies.
A702 (B)	□	Fraction	Medium (10-99)	Hypothesized value varies.
A703 (D)	□	Fraction	Medium (10-99)	Hypothesized value varies.
A704 (E)	□	Fraction	High (>100)	Hypothesized value 1/4.
A705 (F)	□	Fraction	Medium (10-99)	Hypothesized value 1/8.
A706 (H)	□	Fraction	Medium (10-99)	Hypothesized value 1/16.
A707 (J)	□	Fraction	High (>100)	Hypothesized value 1/2.
A708 (K)	□	Fraction	Medium (10-99)	Hypothesized value varies.
A709 (L)	□	Fraction	Medium (10-99)	Base of the 'L' series.
...	...	...	...	*Additional ~21 fractional signs*

Sign_ID	Glyph	Sign_Category	Total_Frequency	Notes
				exist (A709²-A743), including composite fractions.

Note: The full list of 374 signs, including all low-frequency simple signs and the complete set of complex ligatures, is extensive. The table above presents the most frequent and functionally significant signs, with summaries for the remaining categories.

Section 3: Syllabograms: Positional and Contextual Analysis

3.1 Overview of the Syllabary

The phonetic component of Linear A is composed of a core syllabary of approximately 90 signs that appear with sufficient frequency to be considered part of the standard system. These syllabograms represent open syllables (V, CV). The conventional phonetic values assigned to many of these signs are derived from the homomorphy-homophony principle: if a Linear A sign has an identical or near-identical shape to a sign in the deciphered Linear B script, it is hypothesized to have a similar phonetic value. While this principle is a foundational tool for analysis, it remains a working hypothesis, as the underlying Minoan language had a different phonological system from Mycenaean Greek.

3.2 Positional Frequency Analysis

The function of a syllabogram within the Minoan language can be partially inferred from its positional distribution within sign-groups (words). Signs that consistently appear at the end of words are strong candidates for grammatical morphemes (e.g., case endings), while those appearing in initial or medial positions are more likely to be part of lexical roots. This positional data is therefore critical for any structural analysis of the script.

The statistical analysis of the corpus reveals significant differences in sign frequencies and positional preferences between the large archive at Hagia Triada and the combined corpus from all other sites. This discrepancy points to a fundamental question about the Minoan world. It could be evidence of regional linguistic variation, suggesting the existence of distinct dialects of the Minoan language. Alternatively, it may reflect functional specialization between administrative centers; the scribes at Hagia Triada might have been recording different types of goods or using different scribal conventions, leading to a unique statistical profile for their texts. The corpus data provided here, which includes find-spot metadata for inscriptions, allows for a direct investigation of this ambiguity. By analyzing the subsets of data from Hagia Triada and other sites independently, it may be possible to determine whether the structural differences are profound enough to suggest dialectal divergence or are more consistent with variations in administrative practice.

Table 2: Positional Frequencies of Core Syllabograms in Major Corpus Subsets (Data derived from the analysis of D. Packard)

Sign_ID	Conv. Value	Corpus_Subset	Total Occurrences	Initial_Freq_%	Medial_Freq_%	Final_Freq_%
AB53	RA	Hagia Triada	42	11.9%	35.7%	40.5%
AB53	RA	All Others	52	9.6%	61.5%	13.5%
AB59	TA	Hagia Triada	42	14.3%	38.1%	35.7%
AB59	TA	All Others	74	8.1%	21.6%	20.3%
AB06	NA	Hagia Triada	36	13.9%	27.8%	52.8%
AB06	NA	All Others	39	2.6%	64.1%	23.1%
AB08	A	Hagia Triada	26	65.4%	46.2%	11.5%
AB08	A	All Others	47	68.1%	12.8%	6.4%
AB28	I	Hagia Triada	29	17.2%	48.3%	27.6%
AB28	I	All Others	33	51.5%	36.4%	9.1%
AB81	KU	Hagia Triada	41	43.9%	29.3%	12.2%
AB81	KU	All Others	98	18.4%	12.2%	5.1%
AB02	RO	Hagia Triada	36	13.9%	27.8%	52.8%
AB02	RO	All Others	26	7.7%	38.5%	73.1%

Section 4: Logograms and Complex Signs: Semantic and Artifactual Context

4.1 The Logographic System

The majority of Linear A inscriptions are administrative records, and their core components are logograms—signs representing commodities, livestock, people, and manufactured goods. This system includes both simple signs (often syllabograms repurposed for logographic use) and the approximately 164 complex signs, which are ligatures of multiple simple signs. These complex signs are a hallmark of Linear A's administrative system and are overwhelmingly *hapax legomena*, suggesting they were created ad-hoc to represent highly specific items or transactions.

The function of these signs is determined by their context: they typically appear at the end of a descriptive sign-group and are immediately followed by numerals or fractional signs indicating quantity.

Table 3: Contextual Data for Major Commodity Logograms

Sign_ID	Glyph	Conventional_Name	Semantic_Field	Primary_Context	Common_Artifacts
AB120	□	GRA	Dry Product	Administrative	Clay tablets, Roundels
AB122	□	OLIV	Dry Product	Administrative	Clay tablets, Roundels
AB30	□	FIC	Dry Product	Administrative	Clay tablets, Roundels
A303	□	CYP	Dry Product	Administrative	Clay tablets
AB131a	□	VIN	Liquid	Administrative	Clay tablets, Pithoi
A302	□	OLE	Liquid	Administrative	Clay tablets
AB21	□	OVIS	Livestock	Administrative	Clay tablets,

Sign_ID	Glyph	Conventional_Name	Semantic_Field	Primary_Context	Common_Artifacts
					Sealings
AB22	□	CAP	Livestock	Administrative	Clay tablets, Sealings
AB23	□	□	Livestock	Administrative	Clay tablets, Sealings
AB85	□	SUS	Livestock	Administrative	Clay tablets
AB54	□	TELA	Textile	Administrative	Clay tablets
A400-418	□-□	VAS	Vessel	Administrative	Clay tablets

4.2 The Votive Context and the "Libation Formula"

In stark contrast to the administrative texts, a small but significant portion of the Linear A corpus consists of non-administrative inscriptions found in religious contexts. These inscriptions reveal a strong correlation between the material of the artifact, its function, and the type of script used. Inscriptions on durable, high-value materials like carved stone are almost exclusively votive in nature, whereas inscriptions on ephemeral clay are administrative. This pattern establishes a fundamental principle of Linear A usage:

- **Clay Artifacts** (tablets, roundels, sealings) were used for **Administration**, and their texts are dominated by **Logograms and Numerals**.
- **Stone Artifacts** (libation tables, ladles), found primarily in peak sanctuaries, were used for **Religious/Votive** purposes, and their texts are composed of **Syllabograms** arranged in formulaic sequences.

This clear division implies that any computational analysis that fails to distinguish between these contexts—treating the corpus as a monolithic entity—is methodologically flawed. The most prominent example of votive writing is the "libation formula," a recurring sequence of sign-groups. While the exact wording varies, core components like a-ta-i-*301-wa-ja and ja-sa-sa-ra-me appear consistently across multiple artifacts, suggesting they represent the name of a deity and a key ritual term, respectively. The analysis of these formulaic texts, which are purely syllabic and lack logograms, is essential for understanding the phonology and morphology of the Minoan language, separate from the economic jargon of the administrative tablets.

Section 5: The Numerical and Fractional System

5.1 The Decimal System for Integers

The Minoan system for recording integers is a straightforward decimal (base-10) system, a feature that was understood even before the decipherment of Linear B. It is additive and cumulative, with signs representing powers of ten written in descending order of magnitude. The signs are:

- **Units (1):** Vertical strokes (')
- **Tens (10):** Horizontal strokes or dots (−)
- **Hundreds (100):** Circles (∘)
- **Thousands (1000):** Circles with radiating lines (⊹)

5.2 The Fractional System (Klasmatograms)

Linear A also possesses a sophisticated system for representing fractions, a feature essential for the precise accounting of commodities. This system, composed of signs known as klasmatograms, is also additive and cumulative, with fractional signs typically following a logogram or integer and arranged in descending order of value. The system includes a core set of approximately 17 distinct signs, which are conventionally transcribed with capital letters (e.g., J, E, F). While the exact values of many signs are still debated, a combination of statistical, computational, and typological analysis has led to a consistent set of hypothesized values for the most common fractions.

The study of this system reveals a deeper connection between abstract mathematics and concrete economic practice. There is a direct evolutionary link between the Linear A fractional system and the Linear B system of weights and measures. For instance, the Linear A sign hypothesized to represent 1/10 was repurposed in Linear B to denote a specific unit of capacity for dry goods, which was itself one-tenth of a larger standard measure. This continuity implies that the Minoan fractional signs were not merely abstract numbers but were integral to a standardized system of metrology. The choice of a particular fraction may have been linked to the specific commodity being measured. This opens a powerful avenue for analysis: by correlating the co-occurrence of specific commodity logograms with specific fractional signs within the corpus, it may be possible to reconstruct the underlying Minoan system of weights and measures, transforming the analysis from simple script decipherment to a richer reconstruction of Minoan economic practices.

Table 4: The System of Numerical and Fractional Signs

Sign_ID	Glyph	Conventional_Name	Description / Hypothesized Value	Contextual Metadata
-	'	1	Unit	Component of an integer; counted additively up to 9.
-	‒	10	Ten	Component of an integer; counted additively up to 90.
-	°	100	Hundred	Component of an integer; counted additively up to 900.
-	⌖	1000	Thousand	Component of an integer; counted additively.
A707	□	J	1/2	Highest frequency fraction; appears with various commodities.
A704	□	E	1/4	Second highest frequency; often appears in

Sign_ID	Glyph	Conventional_Name	Description / Hypothesized Value	Contextual Metadata
				combination with J.
A705	□	F	1/8	Part of a binary series (J, E, F, H).
A706	□	H	1/16	Part of a binary series (J, E, F, H).
A702	□	B	1/5 or 1/12	Value highly debated; appears with various commodities.
A709	□	L	Base of L-series	Value debated; modified with strokes to create smaller units.
A709²	□	L2	L / 2	Divisive component of the 'L' series.
A709³	□	L3	L / 3	Divisive component of the 'L' series.
A709⁴	□	L4	L / 4	Divisive component of the 'L' series.
A709⁶	□	L6	L / 6	Divisive component of the 'L' series.

Section 6: Conclusions

This report provides a comprehensive and structured data corpus of the complete Linear A signary, prepared for ingestion into the 9-bit Octeract Cypher framework. The assembled data, totaling approximately 374 distinct signs, encompasses the full range of sign types—syllabograms, simple and complex logograms, numerals, and fractions—and is enriched with the most current data on sign frequencies and contextual usage.

The analysis conducted in preparing this corpus has highlighted several fundamental structural properties of the Linear A script that have direct and critical implications for any computational analysis:

1. **Data Sparsity:** The prevalence of *hapax legomena* and low-frequency signs, especially among the complex logograms, indicates that the script was highly specialized. Any analytical model must be designed to be robust in the face of this sparse, long-tail data distribution.
2. **Functional Duality and Context-Dependency:** The function of a sign is not static; it is determined by its position within an inscription. The distinction between a sign's use as a syllabogram versus a logogram is a primary grammatical feature that must be accounted

for through contextual feature engineering.

3. **Material and Functional Segregation:** The corpus is sharply divided along functional lines that correlate directly with the material of the artifact. Administrative texts (logographic, numeric) appear on clay, while votive texts (syllabic, formulaic) appear on stone. Treating these two sub-corpora as a single, undifferentiated dataset would be a significant methodological error.

By structuring the corpus to reflect these properties—providing complete sign lists, positional frequencies, and detailed semantic, artifactual, and contextual metadata—this report equips the Octeract Cypher framework with the necessary data to move beyond simple pattern matching. The provided dataset enables the investigation of advanced hypotheses, including the potential existence of Minoan dialects, the reconstruction of the Minoan metrological system, and the deep structural differences between administrative and religious language. This corpus, therefore, serves not merely as a list of signs, but as a foundation for a nuanced and multi-layered computational exploration of the Minoan world.

Works cited

1. Linear A - Wikipedia, https://en.wikipedia.org/wiki/Linear_A 2. Linear A: Early Cretan Writing System - ThoughtCo, https://www.thoughtco.com/linear-writing-system-of-the-minoans-171553 3. Decoding Linear A, the Writing System of the Ancient Minoans - Hellenic Studies - Georgia State University, https://hellenicstudies.gsu.edu/2023/02/12/decoding-linear-a-the-writing-system-of-the-ancient-minoans-2/ 4. Minoan linguistic resources: The Linear A Digital Corpus - ACL Anthology, https://aclanthology.org/W15-3715.pdf 5. Minoan Cryptanalysis: Computational Approaches to Deciphering Linear A and Assessing Its Connections with Language Families from the Mediterranean and the Black Sea Areas - MDPI, https://www.mdpi.com/2078-2489/15/2/73 6. (PDF) Minoan linguistic resources: The Linear A Digital Corpus - ResearchGate, https://www.researchgate.net/publication/301449236_Minoan_linguistic_resources_The_Linear_A_Digital_Corpus 7. A New Approach to the Decipherment of Linear A, Stage 2 - Fluxus Editions, https://www.fluxus-editions.fr/gla5-cacc.pdf 8. Linear A | Several, Four, Many - Bertold Brecht, https://severalfourmany.wordpress.com/2009/03/04/linear-a/ 9. Linear A - Online resources - Mnamon, https://mnamon.sns.it/index.php?page=Risorse&id=19&lang=en 10. Linear A Texts in phonetic transcription - aegeussociety, https://www.aegeussociety.org/en/useful_websites/linear-a-texts-in-phonetic-transcription/ 11. Cambridge researcher makes steps towards cracking Linear A - Archaeology Wiki, https://www.archaeology.wiki/blog/2021/01/27/cambridge-researcher-makes-steps-towards-cracking-linear-a/ 12. Linear A | Oxford Classical Dictionary, https://oxfordre.com/classics/display/10.1093/acrefore/9780199381135.001.0001/acrefore-9780199381135-e-8927?p=emailAMwmeSz47SQ.o 13. The Signs of Linear A: a palæographical database - SigLA, https://sigla.phis.me/paper.html 14. List of symbols - Linear A - Mnamon, https://mnamon.sns.it/index.php?page=Simboli&id=19&lang=en 15. The Signs of Linear A: a palæographical database 1 Introduction - SigLA, https://sigla.phis.me/paper.pdf 16. The Signs of Linear A: a palæographical database 1 Introduction, https://www.repository.cam.ac.uk/bitstreams/1b5e2fd8-5e8b-4aa3-9f00-40f506bc5949/download 17. Linear A - University of Cambridge, https://www.repository.cam.ac.uk/bitstreams/f6fd5b2a-3c52-4edd-856e-def4e71eb8db/download 18. Linear A fraction signs and their standard transcription. - ResearchGate, https://www.researchgate.net/figure/Linear-A-fraction-signs-and-their-standard-transcription_fig1

_344387974 19. How many Linear A signs do we have? - Minoan language blog, http://minoablog.blogspot.com/2010/05/how-many-linear-signs-do-we-have.html 20. SigLA: The Signs of Linear A - A Palæographical Database - Fluxus Editions, https://www.fluxus-editions.fr/gla5-salg.pdf 21. John G. Younger - Canadian Zen Haiku canadien ISSN 1705-4508, https://linearbknossosmycenae.wordpress.com/tag/john-g-younger/ 22. Does Minoan Linear A Represent One Language?, http://www.anistor.gr/english/enback/p053.htm 23. Linear A - Wikiwand, https://www.wikiwand.com/en/articles/Linear_A 24. Ancient Minoan Civilization: Undeciphered Linear A Script - Brewminate, https://brewminate.com/ancient-minoan-civilization-and-the-mystery-of-the-undeciphered-linear-a-script/ 25. Linear A Script - World History Encyclopedia, https://www.worldhistory.org/Linear_A_Script/ 26. A Semiotic Analysis of Two Linear A Inscribed Ladles - Digital Commons @ IWU, https://digitalcommons.iwu.edu/cgi/viewcontent.cgi?article=1008&context=grs_honproj 27. Chapter one: Theory and methods - ResearchGate, https://www.researchgate.net/publication/286131255_Chapter_one_Theory_and_methods 28. Minoan Stone Vessels with Linear A Inscriptions - aegeussociety, https://www.aegeussociety.org/en/new_book/minoan-stone-vessels-with-linear-a-inscriptions/ 29. Linear A Libation Formula, Minoan Greek : r/HistoricalLinguistics - Reddit, https://www.reddit.com/r/HistoricalLinguistics/comments/1hq549s/linear_a_libation_formula_minoan_greek/ 30. Some reflections on morphology in the language of the Linear A libation formula - Finnish Syntax, https://www.finnishsyntax.co.uk/wp-content/uploads/sites/3/2022/01/Reflections_on_Morphology_in_the_Language_of_the_Linear_A_Libation_Formula.pdf 31. Multiple Sequence Alignment of Libation Formulae suggest Linear A is Minoan-Greek - viXra.org, https://vixra.org/pdf/2105.0007v2.pdf 32. Linear A - Brill Reference Works, https://referenceworks.brill.com/display/entries/EGLO/SIM-00000490.xml 33. The mathematical values of fraction signs in the Linear A script: A computational, statistical and typological approach - ResearchGate, https://www.researchgate.net/publication/344387974_The_mathematical_values_of_fraction_signs_in_the_Linear_A_script_A_computational_statistical_and_typological_approach 34. New Research Sheds Light on Fraction Signs of Still Undeciphered Minoan Linear A Script, https://www.sci.news/archaeology/fraction-signs-minoan-linear-a-script-08835.html 35. The mathematical values of Linear A fraction signs | ScienceDaily, https://www.sciencedaily.com/releases/2020/09/200908113231.htm 36. A new approach to the interpretation of fraction signs in the Linear A script - Università di Bologna, https://www.unibo.it/en/news-and-events/notice-board/a-new-approach-to-the-interpretation-of-fraction-signs-in-the-linear-a-script 37. The mathematical values of Linear A fraction signs, https://www.archaeology.wiki/blog/2020/09/09/the-mathematical-values-of-linear-a-fraction-signs/